

Lead Conversion Analysis Summary for X Education

May 29, 2025

Project Overview

- **Goal:** Identify key factors influencing lead conversion to optimize X Education's sales strategies.
- **Dataset:** 9,240 leads with 37 variables (lead sources, activities, demographics).
- **Approach:** Data cleaning, feature engineering, predictive modeling (Logistic Regression), and strategic analysis.

Key Findings

- **Model Performance:** Achieved 93.5% accuracy and 97.1% ROC AUC, indicating strong predictive power.
- **Top Predictors:**
 - **Tags:** “Will revert after reading the email” (6.48 coefficient) and “Closed by Horizzon” (9.56 coefficient) are highly predictive of conversion.
 - **Lead Source:** Welingak Website (4.26 coefficient) significantly boosts conversion probability.
- **Categorical Focus:** Optimize processes around high-converting tags and enhance Welingak Website's user experience.
- **Strategic Adjustments:**
 - **Aggressive Conversion (Intern Period):** Lower threshold to 0.30–0.35, prioritize recall to maximize true positives.
 - **Minimize Useless Calls (Target Met):** Raise threshold to 0.70–0.75, prioritize precision to focus on high-confidence leads.

Recommendations

1. **Process Analysis:** Replicate successful processes behind high-converting tags.
2. **Channel Optimization:** Invest in Welingak Website and apply its success to other channels.

3. **Threshold Strategies:** Implement dynamic thresholds for aggressive conversion or resource efficiency.
4. **Model Refinement:** Update model with new data and explore additional features.

Business Impact

- Improved resource allocation, higher conversion rates, better ROI on marketing, and data-driven sales strategies.